**Listing 6.13. A thread-safe list with iteration support**

```
template<typename T>
class threadsafe_list
{
    struct node                    <-(1)
    {
        std::mutex m;
        std::shared_ptr<T> data;
        std::unique_ptr<node> next;

        node():                    <-(2)
            next()
        {}

        node(T const& value):      <-(3)
            data(std::make_shared<T>(value))
        {}
    };

    node head;

public:
    threadsafe_list()
    {}

    ~threadsafe_list()
    {
        remove_if([](node const&){return true;});
    }
```